Module 2-1: Linear Regression

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Table of contents

1. Simple Linear Regression

2. Multiple Linear Regression

3. Other Considerations

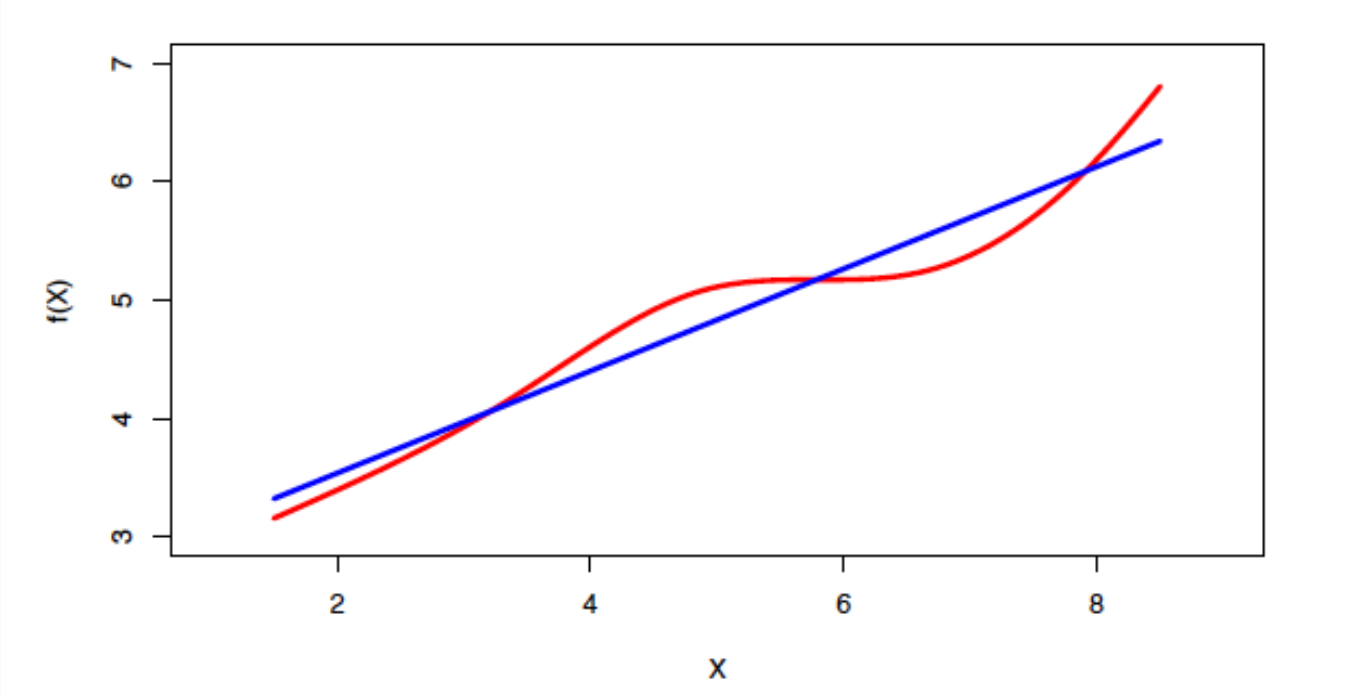
Simple Linear Regression

Linear regression

- Linear regression is a simple approach to supervised learning. It assumes that the dependence of Y on $X_1, \ldots, X_p$ is linear.

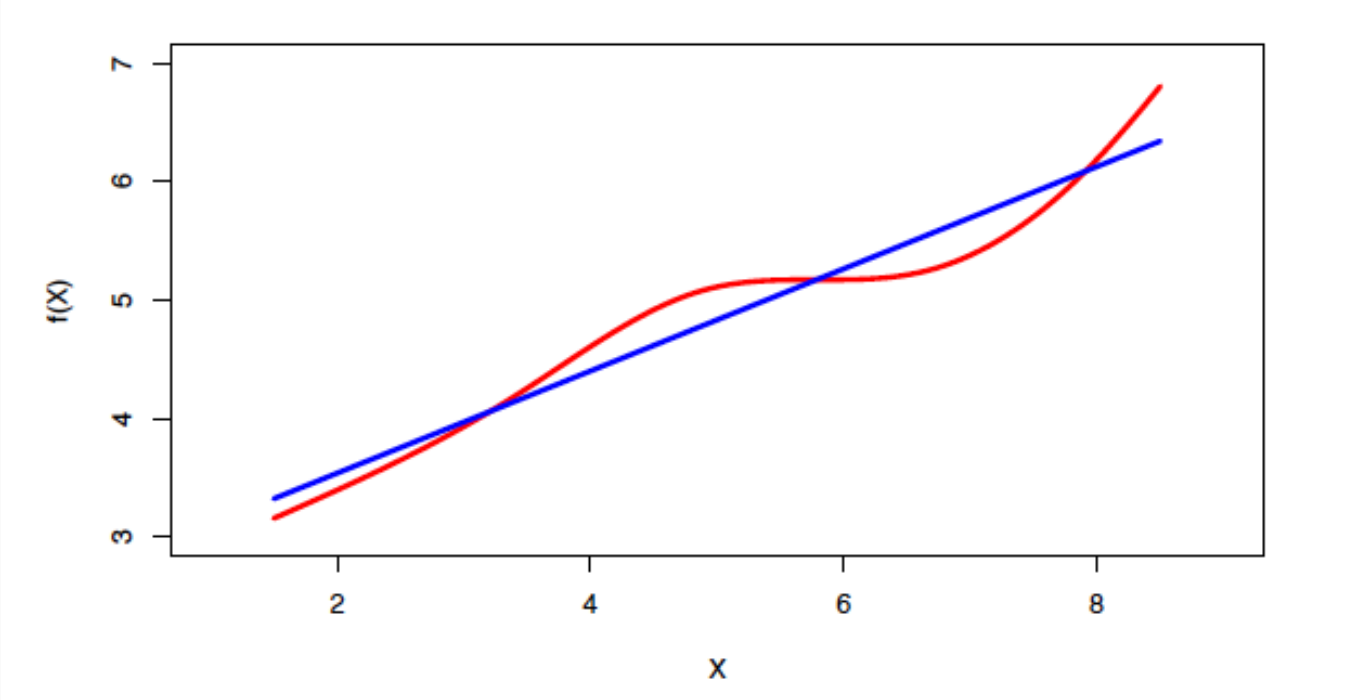
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- True regression functions are never linear!



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- Although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

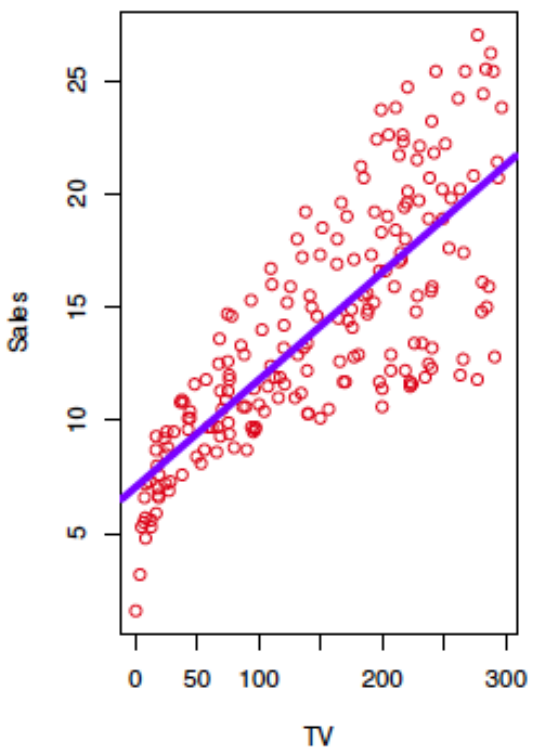
Linear regression for the advertising data

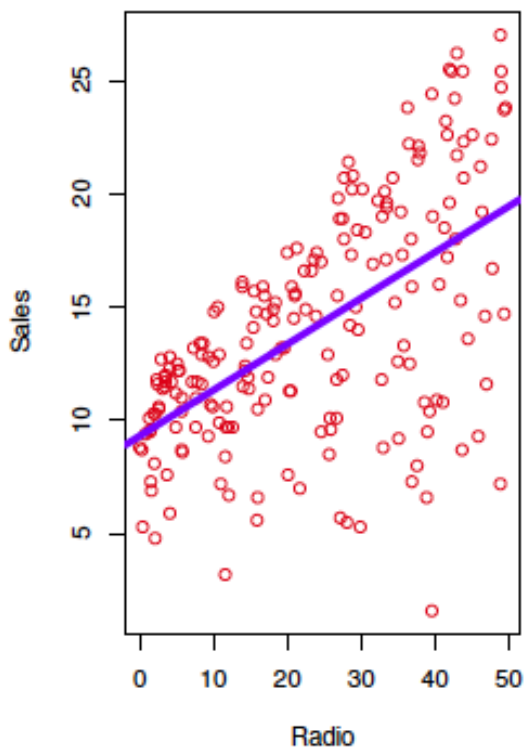
Consider the advertising data shown on the next slide.

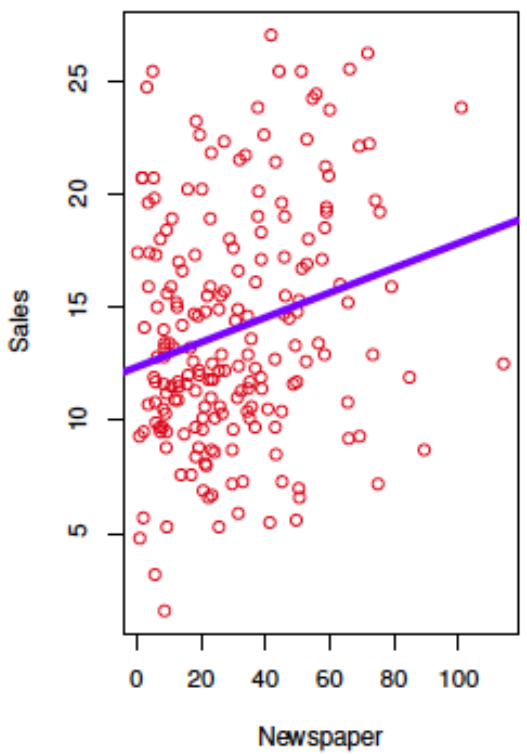
Questions we might ask:

- Is there a relationship between advertising budget and sales?
- How strong is the relationship between advertising budget and sales?
- Which media contribute to sales?
- How accurately can we predict future sales?
- Is the relationship linear?
- Is there synergy among the advertising media?

Advertising data







Simple linear regression

First consider a simple case where $x \in \mathbb{R}$. Data $(x_1, y_1), \ldots, (x_n, y_n)$, where

- x_i is the predictor (independent variable, input, feature)
- y_i is the response (dependent variable, output, outcome)

$$Y = \beta_0 + \beta X + \epsilon$$

We denote the *regression function* as

$$f(x) = \mathbb{E}(Y|X = x).$$

The linear regression model assumes a specific linear form for f,

$$f(x) = \beta_0 + \beta x,$$

which is usually thought of as an approximation to the truth.

Coefficients or parameters — β_0: Intercept; β: Slope

Least squared fitting

- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta} x_i$ are the fitted values

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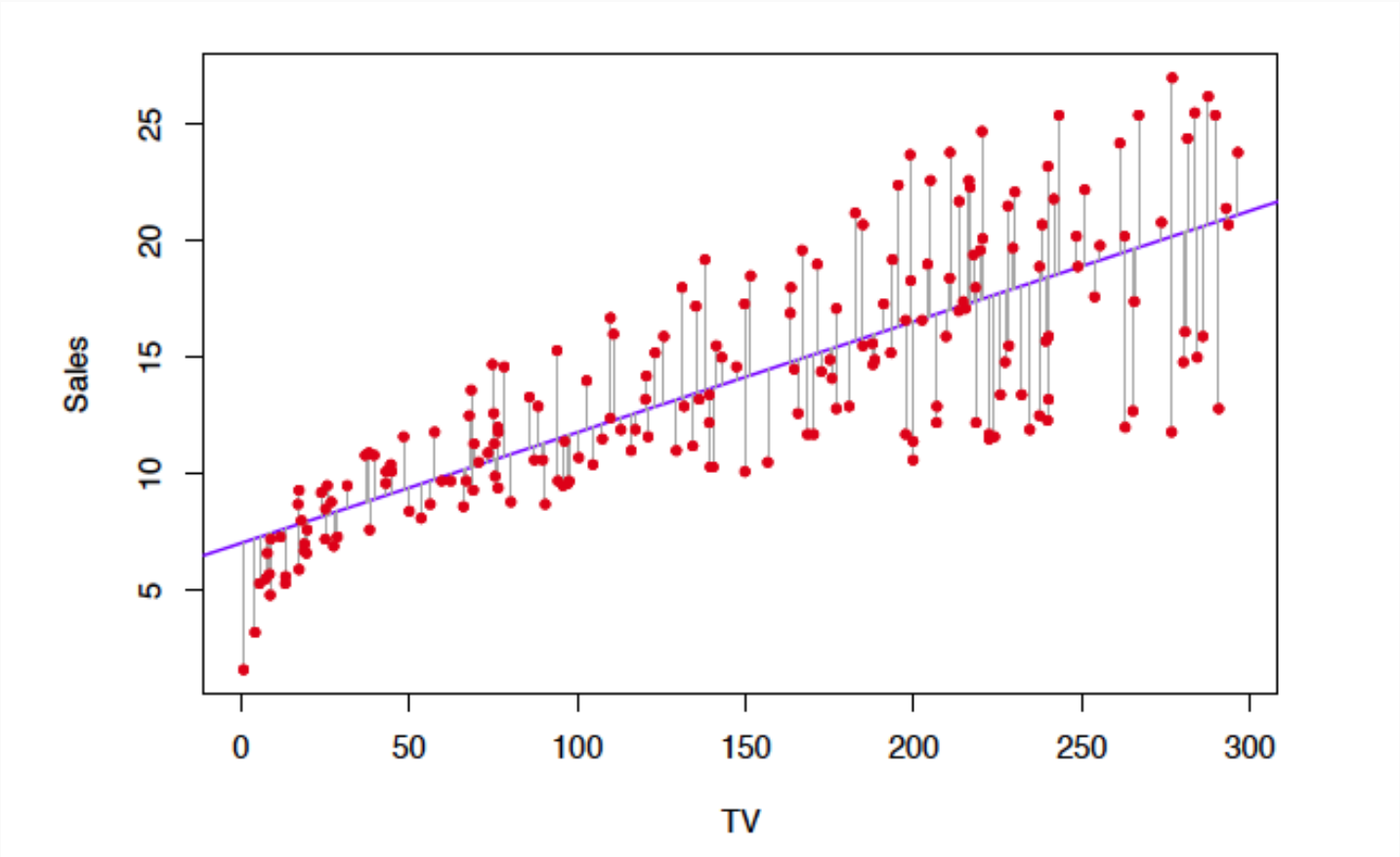
- Minimize:

$$(\hat{\beta}_0, \hat{\beta}) = \underset{\beta_0, \beta}{\operatorname{argmin}} \sum_{i=1}^{n} (y_i - \beta_0 - \beta x_i)^2.$$

- Solution is:

$$\begin{aligned} \hat{\beta} &= \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}, \\ \hat{\beta}_0 &= \bar{y} - \bar{x}\hat{\beta}. \end{aligned}$$

Example: advertising data



The least squares fit for the regression of Sales onto TV. In this case, a linear fit captures the essence of the relationship, although it is somewhat deficient in the left of the plot.

Standard errors and confidence intervals

Assume further that

$$y_i = \beta_0 + \beta x_i + \epsilon_i,$$

where $\mathbb{E}(\epsilon_i) = 0$ and $\mathrm{var}(\epsilon_i) = \sigma^2$. Then

$$se(\hat{\beta}) = \sigma \left(\frac{1}{\sum (x_i - \bar{x})^2} \right)^{1/2},$$

$$se(\hat{\beta}_0) = \sigma \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right)^{1/2},$$

where σ^2 can be estimated by $\hat{\sigma}^2 = \sum (y_i - \hat{y})^2/(n-2)$.

Under additional normality assumption of ϵ_i's, a $(1-\alpha)100\%$ confidence interval of β is

$$\hat{\beta} \pm z_{\alpha/2} \widehat{se}(\hat{\beta}).$$

CI— continued

- That is, there is approximately a 95% chance that the interval

$$\hat{\beta} \pm z_{\alpha/2}\widehat{se}(\hat{\beta})$$

will contain the true value of β under a scenario where we got repeated samples like the present sample.

CI— continued

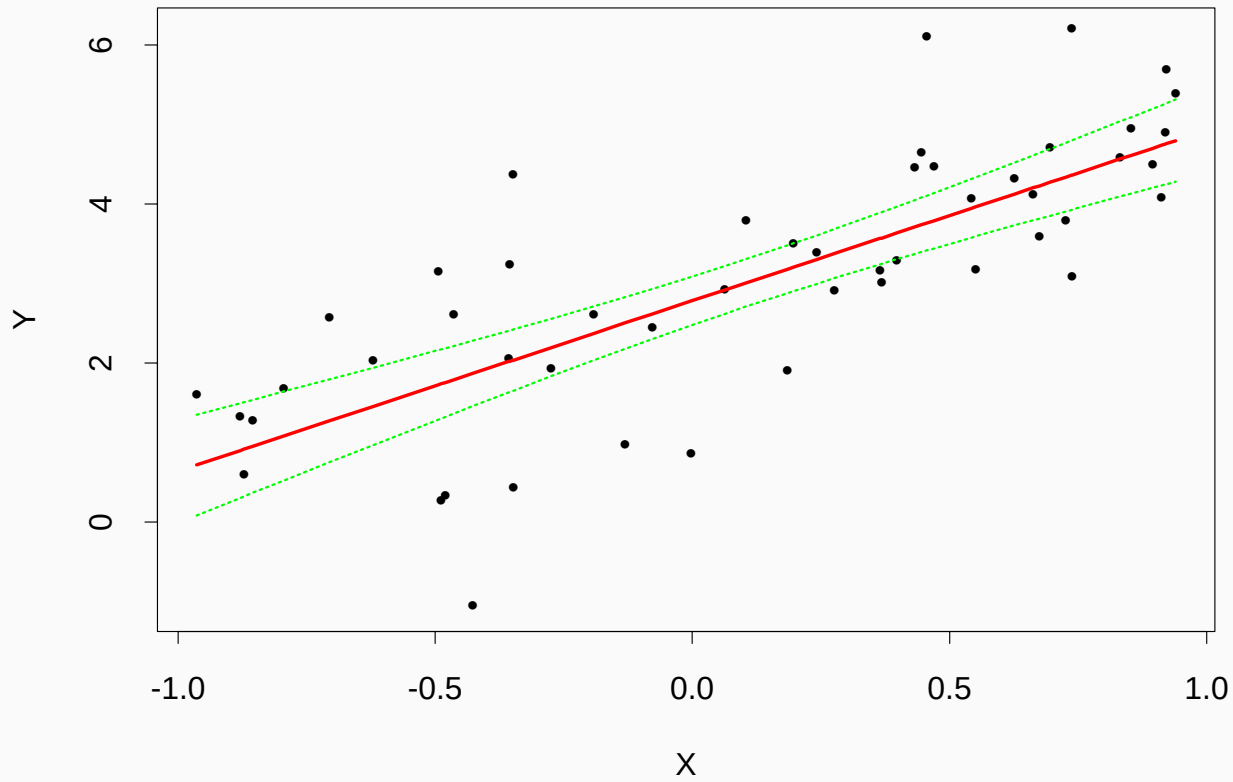
- That is, there is approximately a 95% chance that the interval

$$\hat{\beta} \pm z_{\alpha/2} \widehat{se}(\hat{\beta})$$

  will contain the true value of β under a scenario where we got repeated samples like the present sample.
- For the advertising data, the 95% CI for β is $[0.042, 0.053]$.

Fitted lines

$$\begin{aligned} \hat{f}(x) &= \hat{\beta}_0 + \hat{\beta}x = \bar{y} + \hat{\beta}(x - \bar{x}), \\ se(\hat{f}(x)) &= \left(\mathrm{var}(\bar{y}) + \mathrm{var}(\hat{\beta})(x - \bar{x})^2\right)^{\frac{1}{2}} = \left(\frac{\sigma^2}{n} + \frac{\sigma^2(x - \bar{x})^2}{\sum(x_i - \bar{x})^2}\right)^{\frac{1}{2}}. \end{aligned}$$



Hypothesis testing

- Standard errors can also be used to perform hypothesis tests on the coefficients. The most common hypothesis test involves testing the null hypothesis of

  H_0 : There is no relationship between X and Y versus the alternative hypothesis

  H_A : There is some relationship between X and Y

Hypothesis testing

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  H_0 : There is no relationship between X and Y versus the alternative hypothesis

  $$H_A : \text{There is some relationship between } X \text{ and } Y$$

- Mathematically,

  $$H_0 : \beta = 0 \quad \textit{vs.} \quad H_A : \beta \neq 0,$$

  since if $\beta = 0$ then the model reduces to $Y = \beta_0 + \epsilon$, and X is not associated with Y.

HT —continued

- To test H_0, we compute a t-statistic, given by

$$t = \frac{\hat{\beta} - 0}{se(\hat{\beta})}$$

- This will have a t-distribution with $n - 2$ degrees of freedom, assuming $\beta = 0$
- Using statistical software, it is easy to compute the probability of observing any value equal to $|t|$ or larger. We call this probability the p-value.

Results for the advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

Assessing the overall accuracy of the model

- We compute the Residual Standard Error (RSE)

$$RSE = \sqrt{\frac{1}{n-2}RSS} = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

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- R-squared or fraction of variance explained is

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS},$$

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- It can be shown that in this simple linear regression setting that $R^2 = r^2$, where r is the correlation between X and Y:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{(y_i - \bar{y})^2}}$$

Adverting data results

Quantity	Value
Residual Standard Error	3.26
R^2	0.612
F-statistic	312.1

Multiple Linear Regression

Multiple linear regression

The model is

$$Y = \beta_0 + \beta_1 X_1 + \cdots \beta_{p-1} X_{p-1} + \epsilon$$

$$f(x_i) = \beta_0 + \sum_{j=1}^{p-1} \beta_j x_{ij},$$

or equivalently,

$$\mathbf{f} = \mathbf{X}\,\beta.$$

- $\mathbf{f}$ is n-vector of fitted values
- $\mathbf{X}$ is $n \times p$ matrix of covariates, with ones in the first column
- β is a p-vector of parameters

We interpret β_j as the average effect on Y of a one unit increase in X_j, holding all other predictors fixed. In the advertising example, the model becomes

$$\mathit{Sales} = \beta_0 + \beta_1 \times \mathit{TV} + \beta_2 \times \mathit{radio} + \beta_3 \times \mathit{newspaper} + \epsilon$$

Interpreting regression coefficients

- The ideal scenario is when the predictors are uncorrelated — a balanced design:
  — Each coefficient can be estimated and tested separately.
  — Interpretations such as "a unit change in X_j is associated with a β_j change in Y", while all the other variables stay "fixed", are possible.
- Correlations amongst predictors cause problems:
  — The variance of all coefficients tends to increase, sometimes dramatically
  —Interpretations become hazardous. When X_j changes, everything else changes.
- Claims of causality should be avoided for observational data.

The woes of (interpreting) regression coefficients

"Data Analysis and Regression" by Mosteller and Tukey 1977

- a regression coefficient β_j estimates the expected change in Y per unit change in X_j, with all other predictors held fixed. But predictors usually change together!

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- a regression coefficient β_j estimates the expected change in Y per unit change in X_j, with all other predictors held fixed. But predictors usually change together!
- $Y =$ number of tackles by a football player in a season; W and H are his weight and height. Fitted regression model is $\hat{Y} = b_0 + 0.5W - 0.1H$. How do we interpret $\hat{\beta}_2 < 0$?

Two quotes by famous Statisticians

"Essentially, all models are wrong, but some are useful"
by George Box

"The only way to fnd out what will happen when a complex system is disturbed is to disturb the system, not merely to observe it passively"
by Fred Mosteller and John Tukey, paraphrasing by George Box

Least squared fitting

Minimize:

$$\begin{aligned}\hat{\beta} &= \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p-1}\beta_j x_{ij})^2 \\ &= \underset{\beta}{\operatorname{argmin}} (\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta).\end{aligned}$$

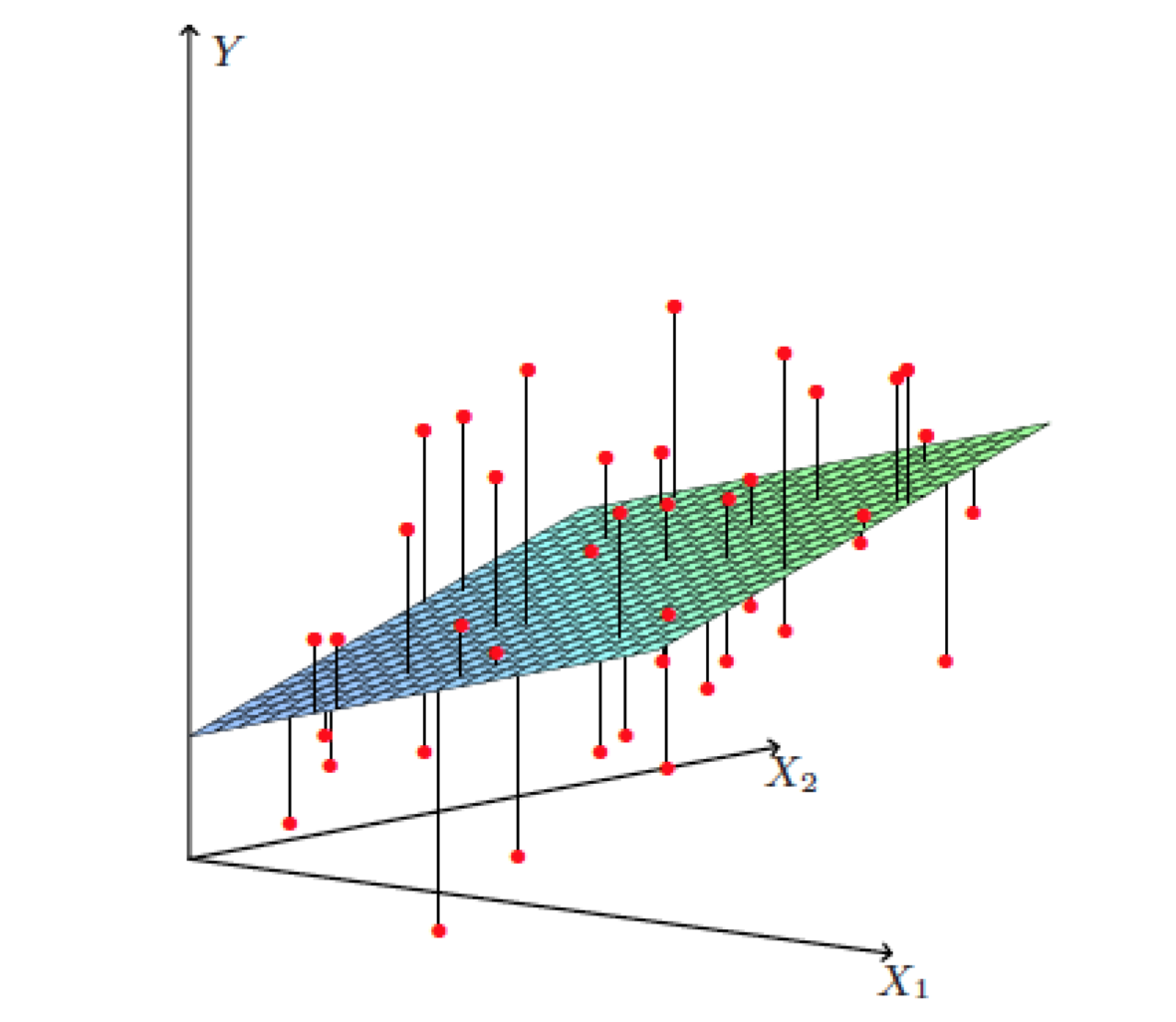
Solution is:

$$\begin{aligned}\hat{\beta} &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}, \\ \hat{\mathbf{y}} &= \mathbf{X}\hat{\beta}.\end{aligned}$$

$$RSS = \sum(y_i - \hat{y}_i)^2$$

Also, $\operatorname{var}(\hat{\beta}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$ and σ^2 is estimated by $\hat{\sigma}^2 = \frac{\|\mathbf{y} - \hat{\mathbf{y}}\|^2}{n-p}$.

An illustrative plot



Results for advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

Correlations:

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

Some important questions

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Some important questions

1. Is at least one of the predictors $X_1, X_2, \ldots, X_{p-1}$ useful in predicting the response?
2. Do all the predictors help to explain Y, or is only a subset of the predictors useful?
3. How well does the model fit the data?
4. Given a set of predictor values, what response value should we predict, and how accurate is our prediction?

Is at least one predictor useful?

For the first question, we can use the F-statistic

$$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)} \sim F_{p,n-p-1}$$

Quantity	Value
Residual Standard Error	1.69
R^2	0.897
F-statistic	570

The Bias-variance tradeoff

A good measure of the quality of $\hat{f}(x)$ is the mean squared error. Let $f_0(x)$ be the true value of $f(x)$ at the point x, then

$$\mathrm{MSE}(\hat{f}(x)) = \mathbb{E}(\hat{f}(x) - f_0(x))^2.$$

This can be written as

$$\mathrm{MSE}(\hat{f}(x)) = \mathrm{var}(\hat{f}(x)) + (\mathbb{E}(\hat{f}(x)) - f_0(x))^2.$$

Typically, when bias is low, variance will be high and vice-versa. Choosing estimators often involves a tradeoff between bias and variance.

- If the linear model is correct for a given problem, then the least squared estimate $\hat{f}$ is unbiased, and has the lowest variance among all unbiased estimators that are linear functions of $\mathbf{y}$ (Gauss-Markov Theorem).
- But there can be biased estimators with smaller MSE.
  - Generally, by regularizing the estimator in some way, its variance will be reduced; if the corresponding increase in bias is small, this will be worthwhile.
  - Examples of regularization: subset selection (forward, backward, all subsets); ridge regression, lasso
- In reality, models are almost never correct, so there will be an additional model bias between the 'best' linear model and the truth.

Variable subset selection

We retain only a subset of the coefficients and set to zero the remaining coefficients. There are different strategies:

- *Subset selection* finds for each $s \in \{0, 1, 2, \ldots, p\}$ the subset of size s that gives smallest residual sum of squares.
- However we often can't examine all possible models, since they are 2^p of them; for example when $p = 40$ there are over a billion models! Instead we need an automated approach that searches through a subset of them. We discuss two commonly use approaches next.
- Rather than search over all possible subsets, we can seek a good path through them.

Forward selection

- *Forward stepwise selection* starts with the intercept and then sequentially adds into the model a variable that most improves the fit.
- Begin with the null model, a model that contains an intercept but no predictors.
- Fit p simple linear regressions and add to the null model the variable that results in the lowest RSS.
- Add to that model the variable that results in the lowest RSS amongst all two-variable models.
- Continue until some stopping rule is satisfied, for example when all remaining variables have a p-value above some threshold.

Backward selection

- *Backward stepwise selection* starts with the full OLS model, and sequentially deletes variables.
- Start with all variables in the model.
- Remove the variable with the largest p-value. That is, the variable that is the least statistically significant.
- The new $(p-1)$-variable model is fit, and the variable with the largest p-value is removed.
- Continue until a stopping rule is reached. For instance, we may stop when all remaining variables have a significant p-value defined by some significance threshold.
- Hybrid stepwise selection strategies sequentially add in the best variable and delete the least important variable.

Model assessment

Typical objectives:

- Choose a value of a tuning parameter for a modeling technique
- Estimate the prediction performance of a given model

Remarks:

- For both objectives, the best approach is to run the procedure on an independent test set, if one is available.
- If possible, one should use different test data for (1) and (2) above: a *validation set* for (1) and a *test set* for (2).
- Often there is insufficient data to create a separate validation or test set. In this case, *Cross-Validation* is useful.

K-fold cross validation

The most popular method for estimating a tuning parameter λ.

Divide the data into K parts: $\mathcal{A}_1, \ldots, \mathcal{A}_K$ ($K = 2, 5, 10$ or n)

- For each $k = 1, \ldots, K$, fit the model with parameter λ to $\{\mathcal{A}_1, \ldots, \mathcal{A}_{k-1}, \mathcal{A}_{k+1}, \ldots, \mathcal{A}_K\}$ giving $\beta^{-k}(\lambda)$, and compute its prediction error on $\mathcal{A}_k$: $E_k(\lambda) = \sum_{x_i \in \mathcal{A}_k} (y_i - x_i^T \beta^{-k}(\lambda))^2$. This gives the cross-validation error
$$CV(\lambda) = \frac{1}{n} \sum_{k=1}^{K} E_k(\lambda).$$
- Compute $CV(\lambda)$ for many λ's and choose the λ yielding the smallest $CV(\lambda)$.

K-fold cross validation

- Cross-validation is often abbreviated as CV.
- In the subset selection procedure, λ is the subset size
- $\beta^{-k}(\lambda)$ is the coefficient for the best model of size λ, found from the training set that leaves out the k-th part of the data
- $E_k(\lambda)$ is its estimated test error on the k-th part.
- Using K-fold CV, the K test error estimates are averaged to give the final CV estimated test error.
- Note that the best model of size λ will be found from each of the K CV training sets.

Bootstrap

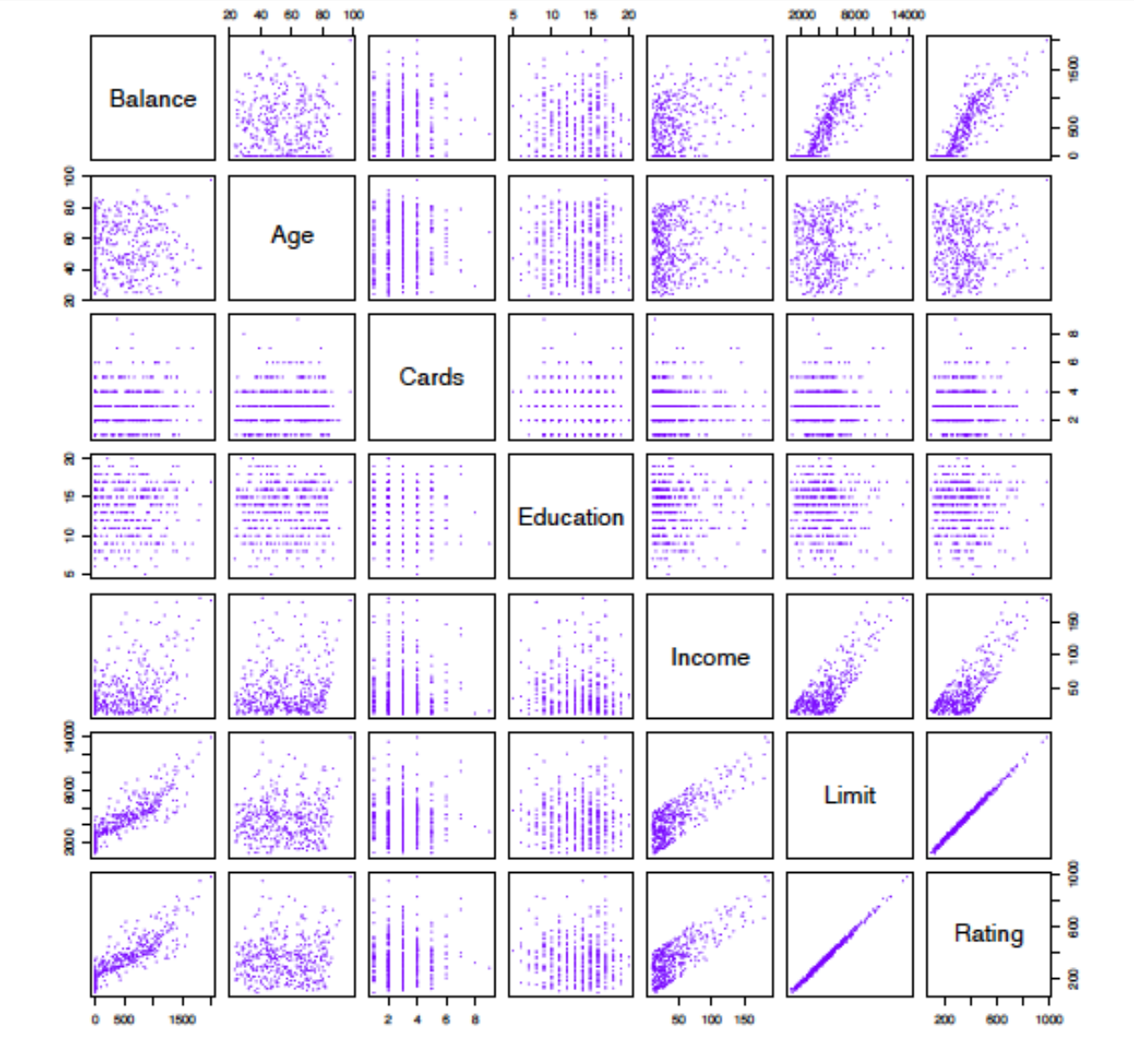
- Bootstrap works by sampling n times with replacement from the training set to form a "bootstrap" data set. Then model is estimated on the bootstrap data set, and predictions are made on the original training set.
- This process is repeated many times and the results are averaged.
- Bootstrap is most useful for estimating standard errors of predictions.
- Can also use modified versions of the bootstrap to estimate prediction error. Sometimes produces better estimates than CV (still an open question!)

Other Considerations

Qualitative Predictors

- Some predictors are not quantitative but are qualitative, taking a discrete set of values.
- These are also called categorical predictors or factor variables.
- See for example the scatterplot matrix of the credit card data in the next slide.
  In addition to the 7 quantitative variables shown, there are four qualitative variables: gender, student (student status), status (marital status), and ethnicity (Caucasian, African American (AA) or Asian).

Credit card data



Qualitative predictors — continued

Example: investigate differences in credit card balance between males and females, ignoring the other variables. We create a new variable

$$x_i = \begin{cases} 1 & \text{if ith person is female} \\ 0 & \text{if ith person is male} \end{cases}$$

Resulting model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if ith person is female} \\ \beta_0 + \epsilon_i & \text{if ith person is male} \end{cases}$$

Interpretation?

Credit card data—continued

Results for gender model:

	Coefficient	Std. Error	t-statistic	p-value
`Intercept`	509.80	33.13	15.389	< 0.0001
`gender[Female]`	19.73	46.05	0.429	0.6690

Qualitative predictors with more than two levels

With more than two levels, we create additional dummy variables. For example, for the ethnicity variable we create two dummy variables. The first could be

$$x_{i1} = \begin{cases} 1 & \text{if ith person is Asian} \\ 0 & \text{if ith person is not Asian} \end{cases}$$

and the second could be

$$x_{i2} = \begin{cases} 1 & \text{if ith person is Causcasian} \\ 0 & \text{if ith person is not Caucasian} \end{cases}$$

Then both of these variables can be used in the regression equation, in order to obtain the model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_2$$

$$= \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if ith person is Asian} \\ \beta_0 + \beta_2 + \epsilon_i & \text{if ith person is Caucasian} \\ \beta_0 + \epsilon_i & \text{if ith person is AA} \end{cases}$$

There will always be one fewer dummy variable than the number of levels. The level with no dummy variable, African American in this example, is known as the baseline.

Credit card data—continued

Results for ethnicity:

	Coefficient	Std. Error	t-statistic	p-value
`Intercept`	531.00	46.32	11.464	< 0.0001
`ethnicity[Asian]`	-18.69	65.02	-0.287	0.7740
`ethnicity[Caucasian]`	-12.50	56.68	-0.221	0.8260

Interactions

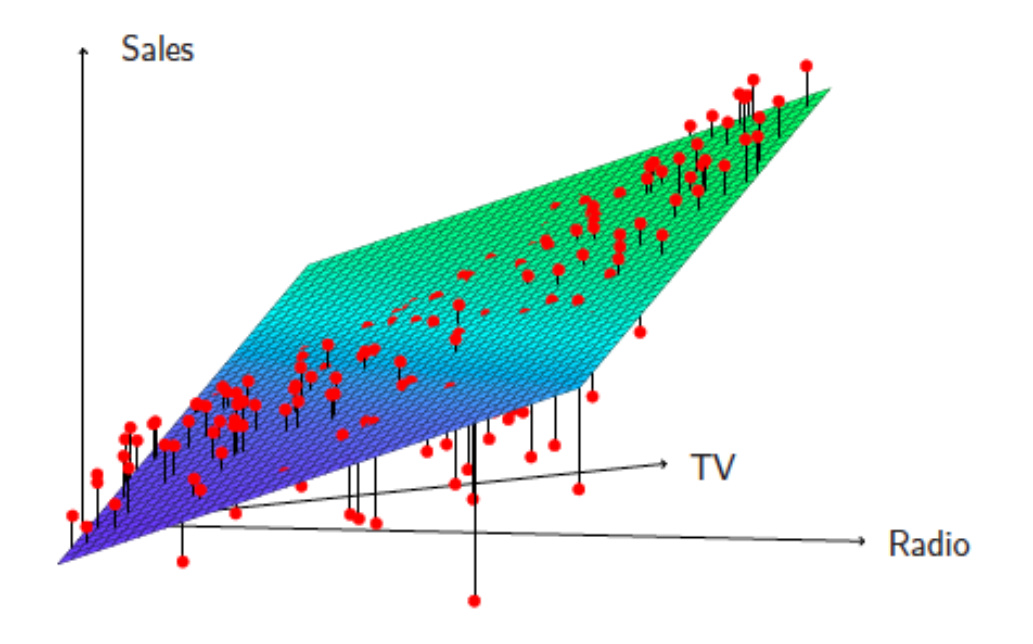
- In our previous analysis of the Advertising data, we assumed that the effect on sales of increasing one advertising medium is independent of the amount spent on the other media.
- For example, the linear model

$$\textit{sales} = \beta_0 + \beta_1 \times \textit{TV} + \beta_2 \times \textit{radio} + \beta_3 \times \textit{newspaper}$$

states that the average effect on sales of a one-unit increase in TV is always β_1, regardless of the amount spend on radio.

- But suppose that spending money on radio advertising actually increases the effectiveness of TV advertising, so that the slope term for TV should increase as radio increases.
- In this situation, given a fixed budget of \$100, 000, spending half on radio and half on TV may increase sales more than allocating the entire amount to either TV or to radio.
- In marketing, this is known as a synergy effect, and in Statistics it is referred to as an interaction effect.

Iteration in the advertising data?



When levels of either TV or radio are low, then the true sales are lower than predicted by the linear model. But when advertising is split between the two media, then the model tends to underestimate sales.

Model taks the form

$$sales = \beta_0 + \beta_1 \times TV + \beta_2 \times radio + \beta_3 \times (radio \times TV) + \epsilon$$

$$= \beta_0 + (\beta_1 + \beta_3 \times radio) \times TV + \beta_2 \times radio + \epsilon$$

Results:

	Coefficient	Std. Error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

Interpretation

- The results in this table suggests that interactions are important.
- The p-value for the interaction term $TV \times radio$ is extremely low, indicating that there is strong evidence for $H_A : \beta_3 \neq 0$.
- The R^2 for the interaction model is 96.8%, compared to only 89.7% for the model that predicts sales using TV and radio without an interaction term.

- This means that $(96.8 - 89.7)/(100 - 89.7) = 69\%$ of the variability in sales that remains after fitting the additive model has been explained by the interaction term.
- The coefficient estimates in the table suggest that an increase in TV advertising of \$1, 000 is associated with increased sales of $(\hat{\beta}_1 + \hat{\beta}_3 \times \textit{radio}) \times 1000 = 19 + 1.1 \times \textit{radio}$ units.
- An increase in radio advertising of \$1, 000 will be associated with an increase in sales of $(\hat{\beta}_2 + \hat{\beta}_3 \times \textit{TV}) \times 1000 = 29 + 1.1 \times \textit{TV}$ units.

Hierarchy

- Sometimes it is the case that an interaction term has a very small p-value, but the associated main effects (in this case, TV and radio) do not.
- The hierarchy principle:
  If we include an interaction in a model, we should also include the main effects, even if the p-values associated with their coefficients are not significant.

- The rationale for this principle is that interactions are hard to interpret in a model without main effects — their meaning is changed.
- Specifically, the interaction terms also contain main effects, if the model has no main effect terms.

Generalizations of the Linear Model

In much of the rest of this course, we discuss methods that expand the scope of linear models and how they are fit:

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- Classification problems: logistic regression, support vector machines
- Non-linearity: kernel smoothing, splines and generalized additive models; nearest neighbor methods.
- Interactions: Tree-based methods, bagging, random forests and boosting (these also capture non-linearities)
- Regularized fitting: Ridge regression and lasso

Additional readings

Fan, J. and Li, R. (2001). Variable Selection via Nonconcave Penalized Likelihood and its Oracle Properties. *J. Amer. Statist. Assoc.*, **96**, 1348-1360.

Efron, B., Hastie, T., Johnstone, I. and Tibshirani, R. (2004). Least angle regression. *Ann. Statist.*, **32**, 407-499.